Zhixu Duan

University of Electronic Science and Technology of China, Chengdu, China, 611731
Homepage: zxuuuustupid.github.io Email: zhixuduan@std.uestc.edu.cn Phone: 15234230092
Google Scholar: Zhixu Duan GitHub: zxuuuustupid ORCID: 0009-0009-6028-5515

Education

- **University of Electronic Science and Technology of China** — Chengdu, China
  Bachelor of Mechanical Design, Manufacturing and Automation, School of Mechanical and Electrical Engineering — *Sep. 2023 - Present*

GPA: 3.97/4.00; Weighted Average: 90.7/100; Major Ranking: 1/83; CET-6: 512

Research Experience

- **Shanghai Jiao Tong University, School of Artificial Intelligence** — Shanghai, China
  Research Assistant at EPICLab; advised by Prof. Linfeng Zhang; efficient and precision intelligent computing — *Nov. 2025 - Present*
- **University of Electronic Science and Technology of China** — Chengdu, China
  Research Assistant at ReliaLab / Center for System Reliability and Safety; advised by Prof. Zuoyi Chen and Prof. Hong-Zhong Huang — *Jan. 2024 - Present*

Research Interests

My research interests focus on AI, robotics, automation and mechanical engineering related topics.

Publications and Projects

(* Equal contribution / first-student author; † corresponding author)

[1] ***Decoupling Intrinsic Fault Features from Domain Variations via Domain-Attribute Fusion for Unseen-Domain Fault Diagnosis***
Accepted by **Advanced Engineering Informatics** (**SCI/CAS Q1**, IF: 9.9); patent filed
Zhixu Duan, et al.

[2] ***Pseudo-fault Data Enhanced Relation Network for Fault Detection and Localization in Train Transmission Systems***
Accepted by **Engineering Applications of Artificial Intelligence** (**SCI/CAS Q1**, IF: 7.5)
Zhixu Duan, Ruoxi Liu, Zuoyi Chen, Hong-Zhong Huang†

[3] ***Unified Health Domain Relation Learning for Train Transmission Systems Fault Detection under Complex Operating Conditions***
Accepted by **Structural Health Monitoring** (**SCI/CAS Q2**, IF: 5.7); patent filed
Zuoyi Chen*, **Zhixu Duan***, Hong-Zhong Huang†

[4] ***Parallel Relation Network for Intelligent Fault Detection and Localization of Train Transmission Systems with Zero-fault Sample***
IEEE PHM 2024 Oral Presentation
Zhixu Duan, Ruoxi Liu, Zuoyi Chen, Hong-Zhong Huang†

[5] ***Reinforcing Cross-Domain Few-Shot Fault Diagnosis of Train Transmission Systems via Reducing Intra-Class and Maximizing Inter-Class Variations***
Accepted by **Advanced Engineering Informatics** (**SCI/CAS Q1**)
Ruoxi Liu, **Zhixu Duan**, Zuoyi Chen, Hong-Zhong Huang†

[6] ***Collaborative Teacher-Student Learning: Simulated Domain Attacks for Class-Intrinsic Feature Learning in Multi-Domain Generalized Fault Diagnosis***
Submitted to **IEEE Transactions on Industrial Informatics** (**SCI/CAS Q1**, IF: 9.9); patent filed
Zhixu Duan, et al.

[7] ***Open-Set Fault Diagnosis Using CLIP with Forward-Reverse Reasoning***
Minor revision at **Computers in Industry** (**SCI/CAS Q1**)
Zuoyi Chen*, **Zhixu Duan***, Hong-Zhong Huang†

[8] ***IndustryCode: A Benchmark for Industry Code Generation***
Submitted to **Neural Information Processing Systems (NeurIPS)**
Contributor

Honors and Awards

- National Scholarship, 2025; National Scholarship, 2024; two-time recipient, top 1.8–1.9%.
- First-Class Outstanding Student Scholarship, UESTC, top 10%.
- "Mechanical and Electrical Elite", selected as one of 10 students across undergraduates, master's, and PhD students in SMEE.
- Principal leader of a National Innovation and Entrepreneurship Program for College Students; completed with excellent conclusion.

Leadership, Outreach, and Skills

- Founding Core Member and Vice President, UESTC Interdisciplinary Association; independently built and maintained the official website; built an advisory board with 5 professors including Prof. Dezhong Yao and Prof. Pedro Antonio Valdes-Sosa. — Feb. 2025 - Present
- Co-hosted the UESTC-IA × DP Technology Uni-Lab Developers' Offline Workshop and led a delegation to Yibin; interviewed by SMEE Idol in "Take Each Step Steadily, and the Distance Will Unfold". — 2025
- Student Mentor for SMEE 2025 freshmen. — Sep. 2025 - Present
- Skills: Python, PyTorch, Jupyter, C, MATLAB, Visio, Origin, PowerPoint, Git, Linux, SolidWorks.